NIH DATA SCIENCE & AI TAXONOMY

Impact of AI on Biomedical Research

Where AI shows up in NIH-funded projects, and what those projects produce: publications, clinical trials, and patents

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THE QUESTION

Funders can’t yet see where AI is used across NIH projects, or what it delivers

How much?

AI-related projects and funding over time

Where?

By institute, method, and research area

With what impact?

Publications, citations, trials, and patents

APPROACH

Keywords find candidates; models decide which projects really use AI

1 · Collect

Every NIH project

ExPORTER, FY2016–FY2025: 799,508 applications combined into 246,754 project records

→

2 · Flag

Keyword taxonomy

706 terms in 36 categories flag 50,481 records, missing almost no AI projects

→

3 · Label

Jev, then ChatGPT

Jev labels each flagged record; ChatGPT resolves the 60% Jev is unsure about

→

4 · Validate

Check the labels

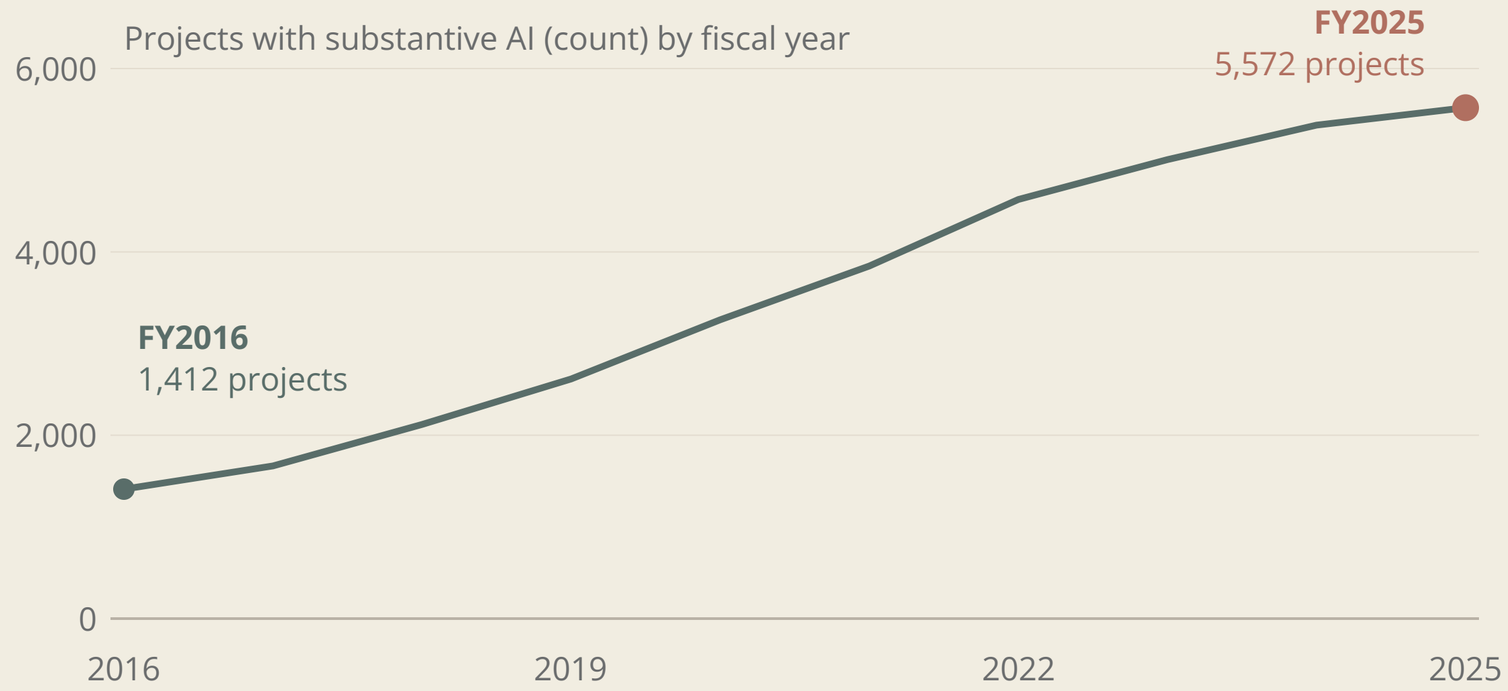
150 coded projects: κ = 0.92 for AI, 0.86 for data science; Claude as reference

Substantive = AI in an aim or the stated approach, not just mentioned · **Jev** = a classification model that returns a confidence for each answer · **κ** = agreement beyond chance (1 = perfect, 0 = chance)

Then linked to publications (ExPORTER), citation impact (iCite, OpenAlex), and patent and trial tables

WHERE AI APPEARS · OVER TIME

Projects using AI grew almost 4× in ten years, to 7.7% of NIH projects



Every year, no single spike

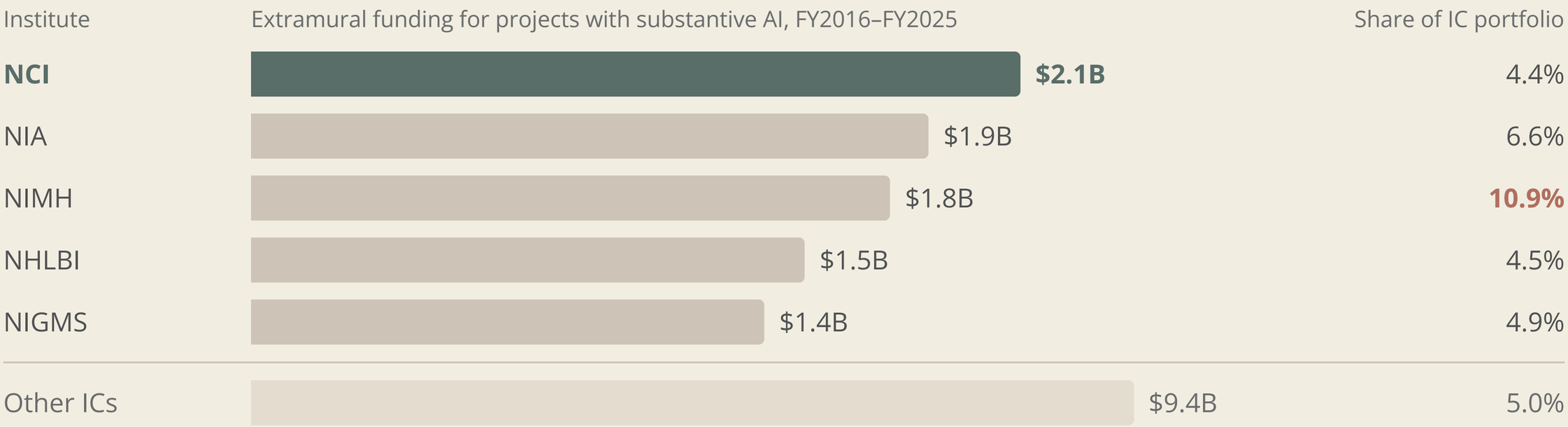
Funding grew faster: $626M to $3.2B a year, 2.2% to 8.5% of extramural funding

Most apply AI: 70% apply it, 21% develop new methods (FY2025)

Substantive AI = AI in an aim or the approach. Counts: all NIH project records active each fiscal year; funding: extramural awards. Source: NIH ExPORTER, FY2016–FY2025; taxonomy labels.

WHERE AI APPEARS · BY INSTITUTE

NCI spends the most on AI; NIMH puts the largest share of its portfolio into it



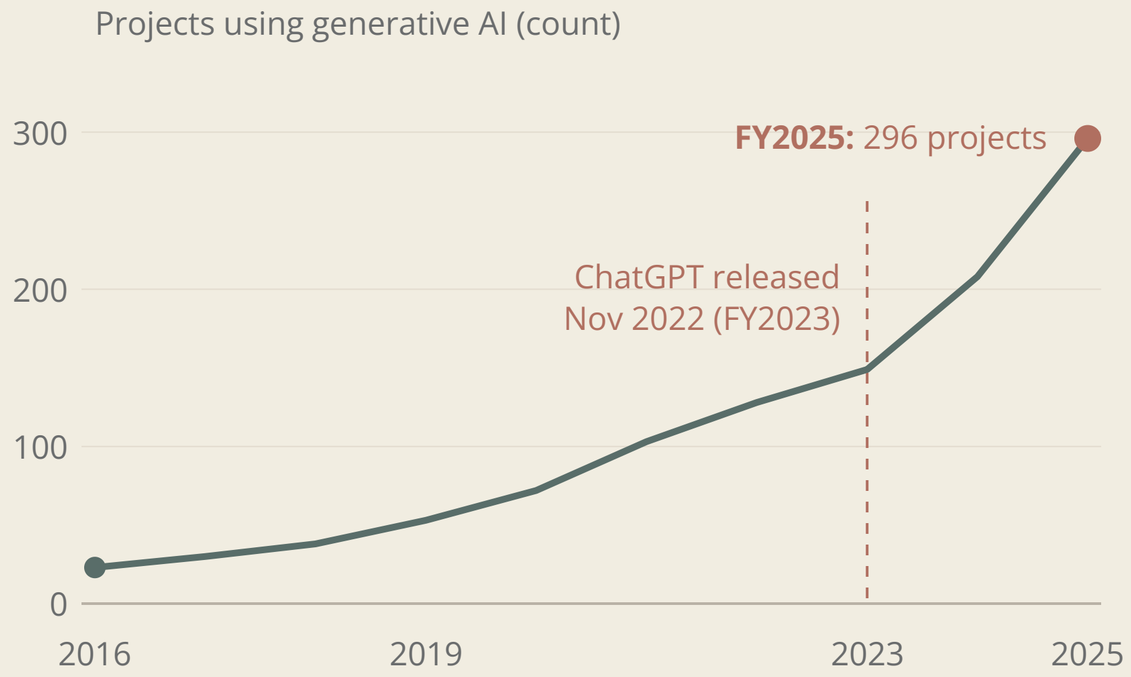
IC = NIH Institute or Center (administering). Funding = extramural awards, subproject costs counted once. Source: NIH ExPORTER; taxonomy labels.

Machine learning is everywhere; vision clusters in imaging, language processing in clinical records

Projects (count)	Imaging	Genomics	Clinical & EHR	Behavioral health	Drug discovery
Machine learning	3,359	3,443	3,828	1,778	838
Deep learning	1,842	1,013	1,309	324	316
NLP	95	164	723	370	38
Computer vision	2,485	543	1,280	250	105
Generative AI	143	101	122	70	78

NLP = natural language processing; EHR = electronic health record. Source: NIH ExPORTER, FY2016–FY2025; taxonomy labels. Low High

Generative AI projects doubled since FY2023, while AI overall grew 11%



What projects do with it (FY2023–FY2025)

Develop new methods **52%**

Apply it to research 40%

Support (data, tools) 7%

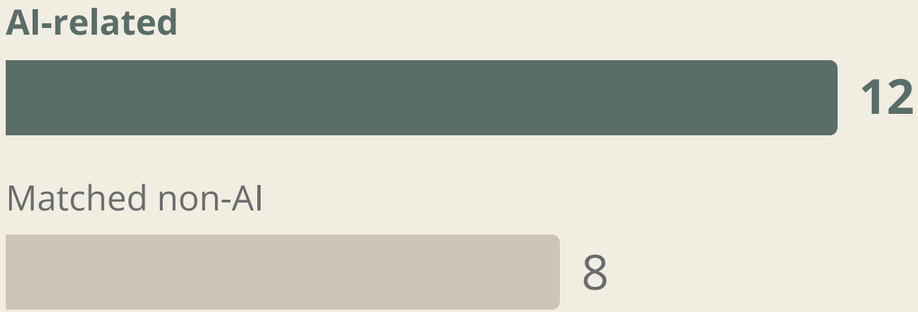
Study AI itself 2%

Top areas: imaging (31%), clinical care and EHR (30%), genomics (26%)

Generative AI includes LLMs (large language models), diffusion models, and earlier generative models such as GANs. Source: NIH ExPORTER; taxonomy labels.

AI-related projects publish more than matched projects: 12 papers vs. 8

Linked papers per project (median)



Citation impact

Citation impact (RCR from iCite, FWCI from OpenAlex) is being added; notebook 13 runs it for the same matched projects.

Matched = same institute, activity code, first fiscal year, and size band; non-AI projects weighted to the AI projects in each group. 91% of AI-related and 89% of matched projects have at least one linked paper.

Source: NIH ExPORTER project and publication link files, FY2016–FY2025; taxonomy labels.

IMPACT · TRANSLATION

Do AI-related projects link to trials and patents more often? Results pending

Projects linked to a clinical trial



Projects linked to a patent



Linked = the project number appears in a ClinicalTrials.gov record or in a patent's government-interest statement (PatentsView). Same matched comparison as the previous slide.

Pending: notebook 13 links ExPORTER patent and clinical study tables to the matched projects. Source when built: ClinicalTrials.gov, PatentsView, NIH RePORTER

SO WHAT

A shared AI taxonomy lets funders plan and report AI investment

1

See it

Track AI investment by institute, method, and area in one view

2

Find gaps

Light cells in the method × area map point to areas to examine

3

Measure impact

Compare outputs and translation against matched projects

Limits: labels agree with hand coding at κ 0.86–0.92; keywords miss 2–6% of data science projects; funding covers extramural awards; recent projects have had little time to publish

Next: add citation impact, patents, and trials; publish the taxonomy; refresh each fiscal year with FY2026